are insects.' Insect 'cuckoos' defy listing; they are so numerous and their habit has been reinvented so often. Some examples that we'll look at have gone beyond familiar cuckooism to fulfil the wildest fantasies that *The Extended Phenotype* might have inspired.

A bird cuckoo deposits her egg and disappears. Some ant cuckoo females make their presence felt in more dramatic fashion. I don't often give Latin names, but *Bothriomyrmex regicidus* and *B. decapitans* tell a story. These two species are both parasites on other species of ants. Among all ants, of course, the young are normally fed not by parents but by workers, so it is workers that any would-be cuckoo must fool or manipulate. A useful first step is to dispose of the workers' own mother with her propensity to produce competing brood. In these two species the parasite queen, all alone, steals into the nest of another ant species. She seeks out the host queen, and rides about on her back while she quietly performs, to quote Edward Wilson's artfully macabre understatement, 'the one act for which she is uniquely specialized: slowly cutting off the head of her victim'. The murderess is then adopted by the orphaned workers, who unsuspectingly tend her eggs and larvae. Some are nurtured into workers themselves, who gradually replace the original species in the nest. Others become queens who fly out to seek pastures new and royal heads yet unsevered.

But sawing off heads is a bit of a chore. Parasites are not accustomed to exerting themselves if they can coerce a stand-in. My favourite character in Wilson's *The Insect Societies* is *Monomorium santschii*. This species, over evolutionary time, has lost its worker caste altogether. The host workers do everything for their parasites, even the most terrible task of all. At the behest of the invading parasite queen, they actually perform the deed of murdering their own mother. The usurper doesn't need to use her jaws. She uses mind-control. How she does it is a mystery; she probably employs a chemical, for ant nervous systems are generally highly attuned to them. If her weapon is indeed chemical, then it is as insidious a drug as any known to science. For think what it accomplishes. It floods the brain of the worker ant, grabs the reins of her muscles, woos her from deeply ingrained duties and turns her against her own mother. For ants, matricide is an act of special genetic madness and formidable indeed must be the drug that drives them to it. In the world of the extended phenotype, ask not how an animal's behaviour benefits its genes; ask instead whose genes it is benefiting.